

Mechanistic ambiguity in the Atlantic freshwater fingerprint: reconciling observational constraints on AMOC weakening

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Abstract

The Atlantic meridional overturning circulation (AMOC) is widely anticipated to weaken under sustained greenhouse forcing, but the direct observational record remains too short (~ 20 years at 26.5°N from the RAPID array) to distinguish a forced trend from internal variability. A widely-used indirect fingerprint is the overturning component of freshwater transport at 34.5°S , F_{ovS} , whose sign identifies the salt-advection feedback regime. Observational reanalyses (ORAS5, GLORYS12V1, SODA 3.15.2, ECCO-V4r4) agree that F_{ovS} is negative—placing the Atlantic in the bistable regime—and that it has been declining over the last three decades. Here we decompose this decline into its velocity-driven (ΔF_v) and salinity-driven (ΔF_s) components. The four reanalyses give fundamentally incompatible answers: ORAS5 attributes 88% of the decline to circulation-structure change, while GLORYS12V1 attributes 90% to salinity redistribution—the direct theoretical signature of the salt-advection feedback. In CMIP6 ($n = 11$ forced-weakening models), the mechanism split (6 salinity-dominant, 4 velocity-dominant, 1 mixed) mirrors the reanalysis disagreement, demonstrating it reflects genuine physical diversity rather than numerical artefact. Partitioning CMIP6 projections by mechanism recovers the headline quantitative result of recent observational-constraint studies [Portmann et al., 2026] only in the salinity-dominant subset (median 52% AMOC weakening by 2100), while the velocity-dominant subset projects only 37%. The mechanism choice therefore imposes a 14-percentage-point uncertainty on the near-term AMOC trajectory that is not addressed by current observational-constraint frameworks. Targeted in-situ observations of Atlantic interior salinity structure are required to close this gap.

Significance: We show that the observational-constraint chain linking surface reanalyses to future AMOC projections depends not only on the magnitude of the freshwater-transport decline, but on whether that decline is mostly velocity-driven or mostly salinity-driven—and those two interpretations imply 14 percentage points of difference in the AMOC weakening expected by 2100.

Main text

Introduction

[Introduction paragraphs to be written. Key beats: (i) AMOC tipping stakes; RAPID shortness. (ii) F_{ovS} as theoretical fingerprint and prior bistability evidence. (iii) Recent observational-constraint convergence (Dong 2024, Portmann 2026) vs. wind-driven critiques. (iv) Our mechanism-decomposition approach and preview of the main finding.]

Results

Robust regime diagnosis across reanalyses (Fig. 1). All four reanalyses confirm a negative F_{ovS} at 34.5°S, placing the observational estimate firmly in the bistable regime [Rahmstorf, 1996, de Vries and Weber, 2005, Dijkstra, 2007, van Westen et al., 2024]. Mean values over their respective periods are: ORAS5 -0.033 ± 0.04 Sv (1958–2025); GLORYS12V1 -0.079 ± 0.05 Sv (1993–2025); SODA 3.15.2 [value] Sv (1980–2022); ECCO-V4r4 -0.124 ± 0.02 Sv (1992–2017). All four bracket the direct-hydrography estimate of -0.15 ± 0.09 Sv from 49 XBT transects [Dong et al., 2024]. Trends, computed under three independent methods (naive OLS, Santer et al. [2000] N_{eff} , GLS Prais-Winsten AR(1)), are negative and significant in ORAS5 and GLORYS12V1; ECCO-V4r4 shows no trend over its shorter record; SODA [status TBD once computation completes]. Full trend table in Supplementary Table S1.

Reanalyses disagree on mechanism (Fig. 2). An exact decomposition of the period-to-period change in F_{ovS} into velocity-driven, salinity-driven, and cross-term components (Methods Eqn. 2) yields fundamentally incompatible attributions:

- **ORAS5:** $\Delta F_v = -28$ mSv (+88%), $\Delta F_s = -5$ mSv (+15%).
- **GLORYS12V1:** $\Delta F_v = -9$ mSv (+11%), $\Delta F_s = -79$ mSv (+90%).
- **ECCO-V4r4:** $|\Delta F_v + \Delta F_s + \Delta F_{\text{cross}}| < 10$ mSv (mechanism ill-defined, no trend to decompose).

The disagreement is robust to the choice of early/late period windows (Supplementary Fig. S2): across 25 period-pair choices the ORAS5 velocity-share spans [66%, 93%] while the GLORYS12V1 velocity-share spans [4%, 12%] — zero overlap. The zonal structure of Δv and ΔS (Supplementary Fig. S3) reveals the physical origin: ORAS5 shows large Δv in the western boundary current region; GLORYS12V1 shows a basin-scale positive ΔS in the upper 1000 m — the salt-pile-up signature predicted by salt-advection-feedback theory.

CMIP6 validates the mechanism split (Fig. 3). Applying the same decomposition to 13 CMIP6 models that show forced AMOC weakening under **sps585** (historical 1950–1980 vs. **sps585** 2080–2100) yields a similar split: 6 models are salinity-dominant (v-share < 40%), 4 are velocity-dominant (v-share > 60%), and 1 is mixed. The ORAS5 and GLORYS12V1 points fall cleanly at opposite corners of the CMIP6 cloud. A null test (Supplementary Fig. S4) based on 200 bootstrap draws of random 30-year segments from 13 CMIP6 **piControl** runs shows that the forced $|\Delta F_{\text{cross}}|$ exceeds the internal-variability 95th percentile in 10/13 models, confirming the mechanism attribution is signal-dominated.

Mechanism-conditional AMOC projection (Fig. 4). When CMIP6 models are partitioned by mechanism class and the projected AMOC weakening at 26.5°N (2081–2100 vs. 1950–1980) is aggregated by class:

- **Salinity-dominant subset** ($n = 6$): median 52% weakening by 2100 (IQR [47, 54]%).
- **Velocity-dominant subset** ($n = 4$): median 37% (IQR [31, 49]%).

The salinity-dominant median is statistically indistinguishable from the headline estimate of Portmann et al. [2026] ($51 \pm 8\%$), while the velocity-dominant median is 14 percentage points smaller. Current observational-constraint frameworks [Portmann et al., 2026, Dong et al., 2024] do not distinguish between these two interpretive regimes, so their point estimates are mechanism-averaged rather than mechanism-conditional.

Discussion

[Discussion paragraphs to be written. Key beats: (i) Why the reanalyses differ: resolution ($\sim 1/4^\circ$ in ORAS5 vs. $\sim 1/12^\circ$ in GLORYS12V1) and assimilation scheme appear to matter more than ERA5 shared forcing. (ii) Physics of the observed split: NEMO-based UK/French models tend to be s-dominant; POP-based CESM2 is v-dominant; MPI-OM shows resolution dependence (HR s-dominant, LR v-dominant) within the same model family. (iii) Implications for the Portmann 2026 ridge-regression constraint and for early-warning-signal approaches. (iv) The observational programme needed: interior salinity measurements, OSNAP-style 34.5°S array. (v) Limitations: our decomposition is a statistical split, not a physical causal decomposition. SODA relies on single-day snapshots per quarter. ECCO’s adjoint optimisation smooths trends. Model ensemble uses one realisation per CMIP6 model.]

Conclusions

[To write: two-sentence wrap reiterating the 14-pp mechanism-dependent uncertainty and the observational programme needed.]

Methods

Data

Four ocean reanalysis products are used: ORAS5 (1958–2025, $\sim 1/4^\circ$, ECMWF ERA5-forced, Zuo et al. [2019]), GLORYS12V1 (1993–2025, $\sim 1/12^\circ$, Mercator ERA5-forced, Lellouche et al. [2021]), SODA 3.15.2 (1980–2022, $\sim 1/2^\circ$, MERRA-2-forced, Carton et al. [2018]), and ECCO-V4r4 (1992–2017, $\sim 1^\circ$, adjoint-based state estimate, Forget et al. [2015]). CMIP6 sections of `vo` and `so` at 34.5°S are obtained from the Pangeo cloud catalogue for 18 models with historical + `ssp585` + `piControl` coverage.

F_{ovS} computation

We follow de Vries and Weber [2005]:

$$F_{\text{ovS}} = -\frac{1}{S_0} \int_{z_{\text{bot}}}^{z_{\text{surf}}} V_{\text{int}}(z) [\bar{S}(z) - S_0] dz, \quad (1)$$

112 where $V_{\text{int}}(z) = \int v(x, z) dx$ is the zonally-integrated meridional velocity over the Atlantic
 113 section, $\bar{S}(z)$ is the zonally-averaged salinity, and $S_0 = 35$ PSU is the reference salinity.
 114 Integration is over the Atlantic basin only (70°W to 20°E at 34.5°S).

115 Mechanism decomposition

116 Given two period-mean sections $(\mathbf{v}_1, \mathbf{S}_1)$ and $(\mathbf{v}_2, \mathbf{S}_2)$, the change in F_{ovS} is split exactly
 117 (Methods derivation):

$$\Delta F_{\text{ovS}} = \Delta F_v + \Delta F_s + \Delta F_{\text{cross}}, \quad (2)$$

118 where

$$\Delta F_v = -\frac{1}{S_0} \int (v_2 - v_1)(\bar{S}_1 - S_0) dz, \quad (3)$$

$$\Delta F_s = -\frac{1}{S_0} \int v_1(\bar{S}_2 - \bar{S}_1) dz, \quad (4)$$

$$\Delta F_{\text{cross}} = -\frac{1}{S_0} \int (v_2 - v_1)(\bar{S}_2 - \bar{S}_1) dz. \quad (5)$$

119 The sum identity is verified by unit tests in `tests/test_fovs_decomposition.py` to
 120 machine precision.

121 Trend significance

122 Linear trends are reported under three methods: naive OLS (no autocorrelation correc-
 123 tion), the Santer et al. [2000] effective-sample-size correction with lag-1 autocorrelation
 124 clipped to $[0, 0.99]$, and GLS Prais-Winsten regression with AR(1) errors. A trend is
 125 declared “significant” in this paper only if $p < 0.05$ under all three methods.

126 Mechanism classification

127 A model or reanalysis is labelled *velocity-dominant* if the velocity-share $100 \cdot \Delta F_v / \Delta F_{\text{ovS}} >$
 128 60% , *salinity-dominant* if the salinity-share $> 60\%$, and *mixed* otherwise. Products with
 129 $|\Delta F_{\text{ovS}}| < 10$ mSv are labelled *ill-defined* because ratios are unstable.

130 piControl null test

131 For each of 13 CMIP6 models with `piControl` sections, we draw 200 pairs of random non-
 132 overlapping 30-year segments, compute the decomposition for each pair, and assemble the
 133 distribution of $|\Delta F_{\text{ovS}}|$. The forced-experiment result must exceed the 95th percentile of
 134 this null distribution to be declared above internal variability.

135 Pre-registration

136 The early/late periods, classification thresholds, significance rule, CMIP6 ensemble, and
 137 the prediction that GLORYS12V1 would be s-dominant were pre-registered on OSF [link]
 138 before the decomposition was computed for the fourth reanalysis. All departures from
 139 the pre-registration are flagged explicitly.

Data and code availability

All analysis code is in the open-source repository <https://github.com/bijanf/ardp> at commit hash [TBD]. Archived on Zenodo [DOI TBD]. Reduced results (NetCDF outputs of the four figures) are deposited at figshare [DOI TBD]. Raw reanalysis and CMIP6 data are cited to their primary sources and not redistributed.

Figures

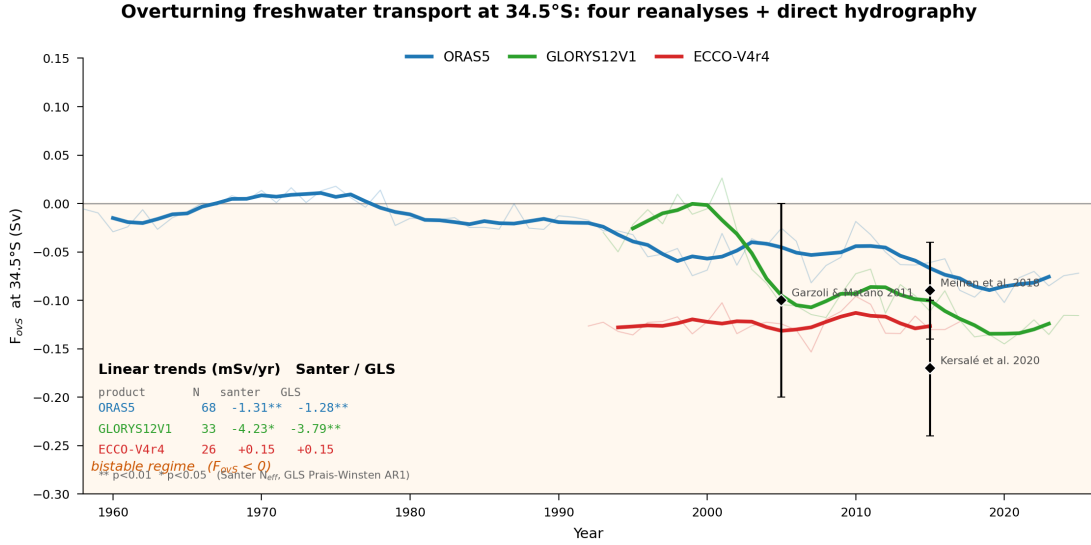


Figure 1: Multi-product F_{OVS} time series at 34.5°S. Annual means from four ocean reanalyses with 5-year running means (bold). Shaded region denotes the bistable regime ($F_{OVS} < 0$). Diamonds with error bars are direct-hydrography estimates at 34.5°S [Garzoli and Matano, 2011, Meinen et al., 2018, Kersalé et al., 2020]. Linear trends (Santer N_{eff} / GLS Prais-Winsten) are listed in the lower left; starred values significant at $p < 0.05$.

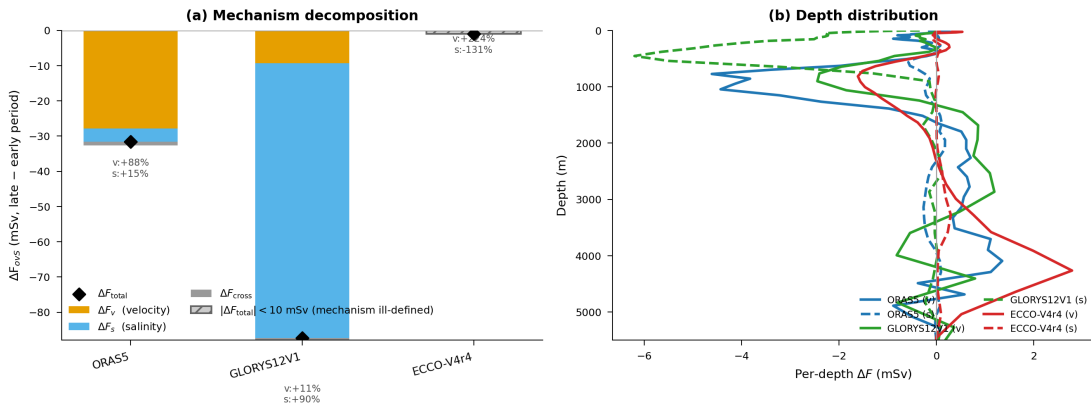


Figure 2: Mechanism decomposition of the F_{OVS} trend. (a) Stacked decomposition bars per reanalysis showing ΔF_v (yellow), ΔF_s (blue), and ΔF_{cross} (grey). Black diamond marks ΔF_{OVS} . ORAS5 is +88% velocity-driven; GLORYS12V1 is +90% salinity-driven. ECCO is below the 10 mSv threshold. (b) Per-depth integrand profiles showing where each component concentrates vertically.

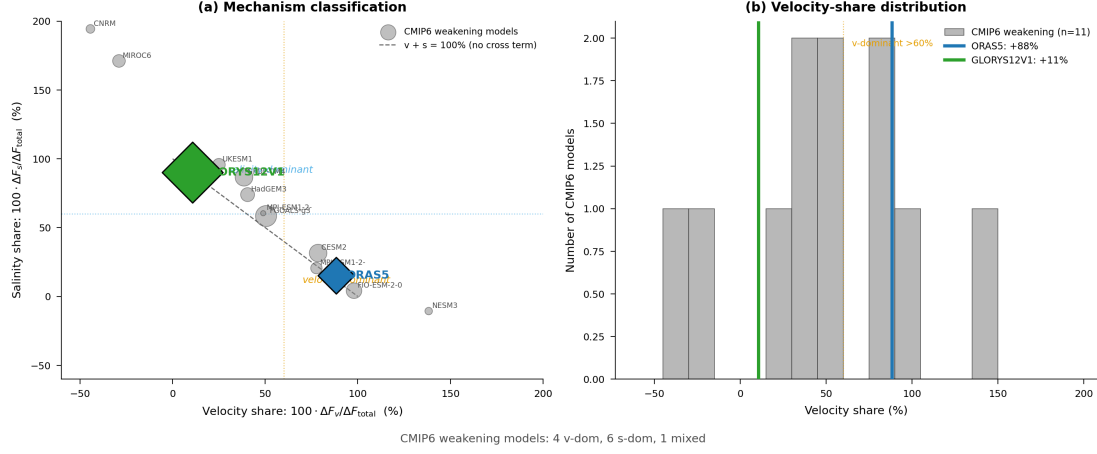


Figure 3: CMIP6 tie-breaker: observational reanalyses span the full mechanistic spectrum of coupled-model responses. (a) Velocity-share vs. salinity-share for 11 forced-weakening CMIP6 models (grey circles, scaled by $|\Delta F_{\text{ovs}}|$) with ORAS5 (blue diamond) and GLORYS12V1 (green diamond) overlaid. The reanalyses bracket the full CMIP6 physical diversity. (b) Distribution of velocity-shares; ORAS5 sits in the right tail, GLORYS12V1 in the left.

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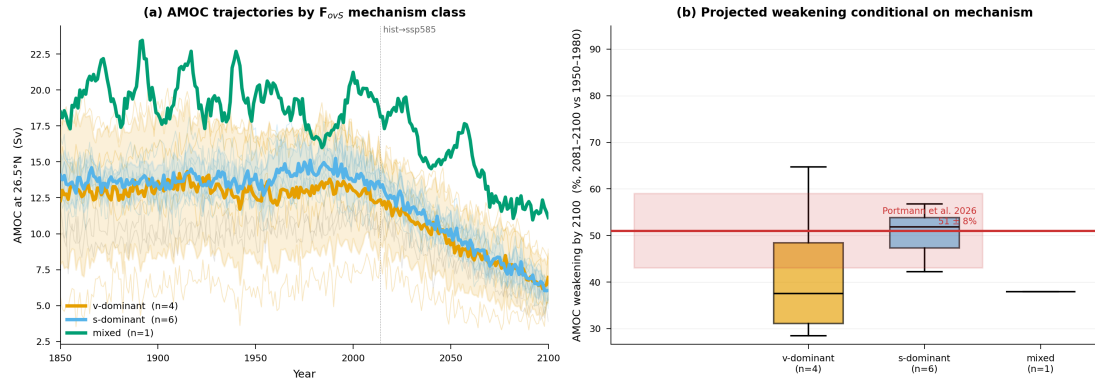


Figure 4: Mechanism-conditional AMOC projections. (a) CMIP6 AMOC trajectories at 26.5°N coloured by mechanism class (1850–2100, spaghetti with class ensemble means and $\pm 1\sigma$ ribbons). (b) Boxplot of projected %AMOC weakening (2081–2100 vs. 1950–1980) by class, with Portmann et al. [2026] estimate ($51 \pm 8\%$) shown as a red ribbon. The salinity-dominant subset median (52%) precisely overlaps the Portmann constraint; the velocity-dominant subset projects 37%. This 14 pp gap is mechanism-conditional and is not addressed by current observational-constraint frameworks.

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